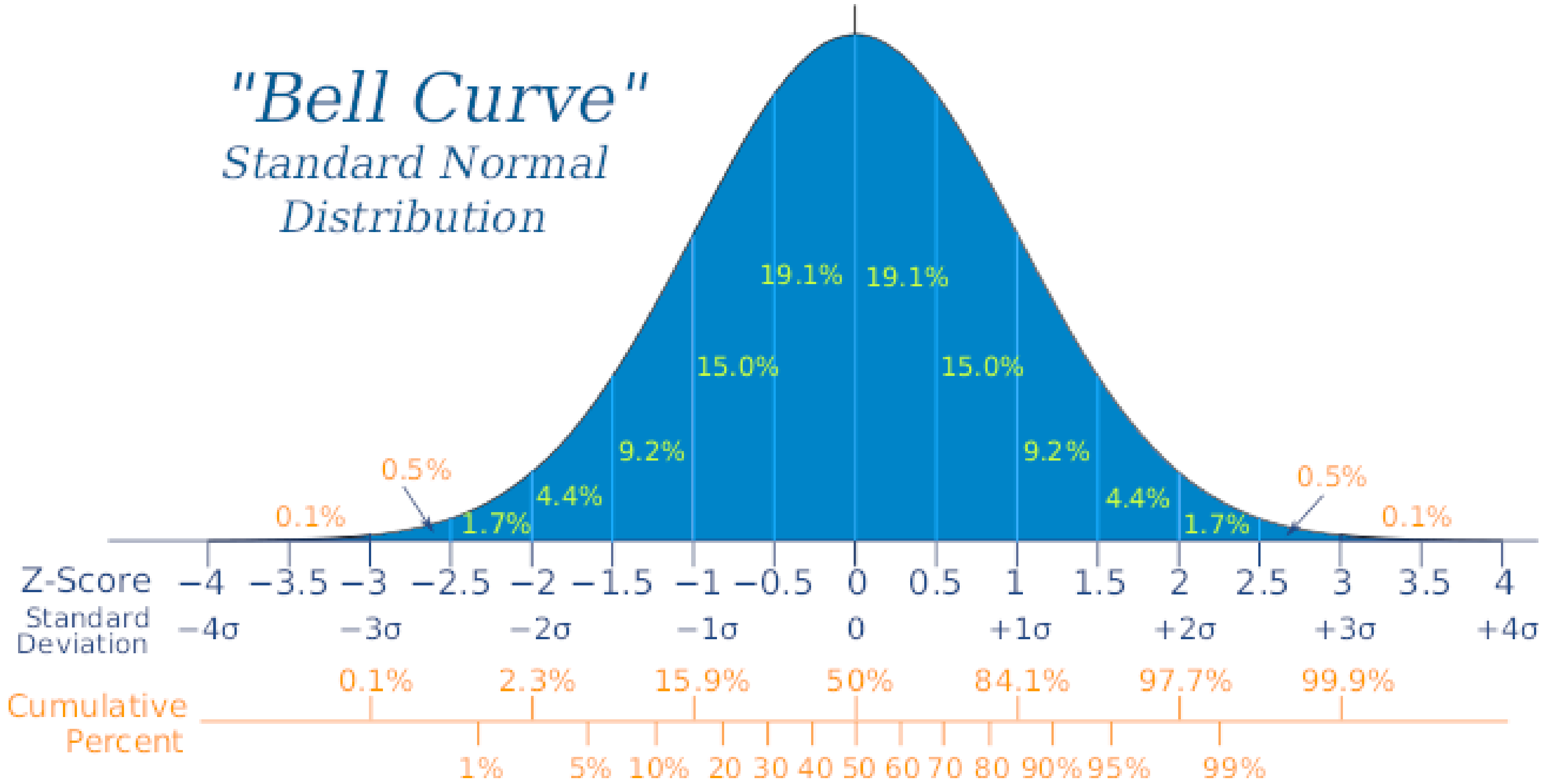
Lecture 05 – Basic RV Distributions: Binomial, Poisson, Gaussian

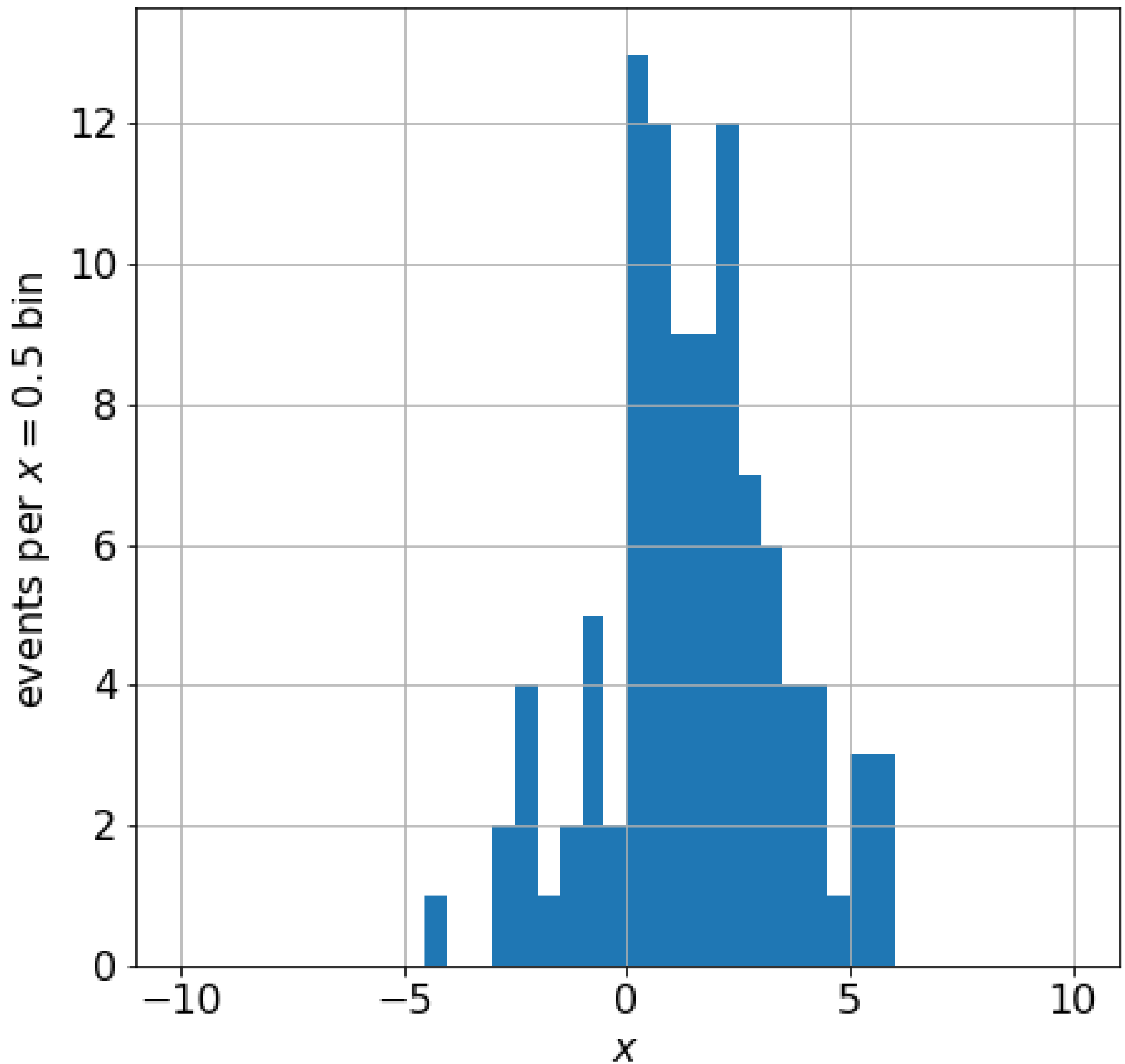


Physics 4730

Fall Semester 2026

PDFs and Empirical PDFs

- If I make n measurements of a random variable x, the results can be sorted into a histogram to determine the experimental distribution.



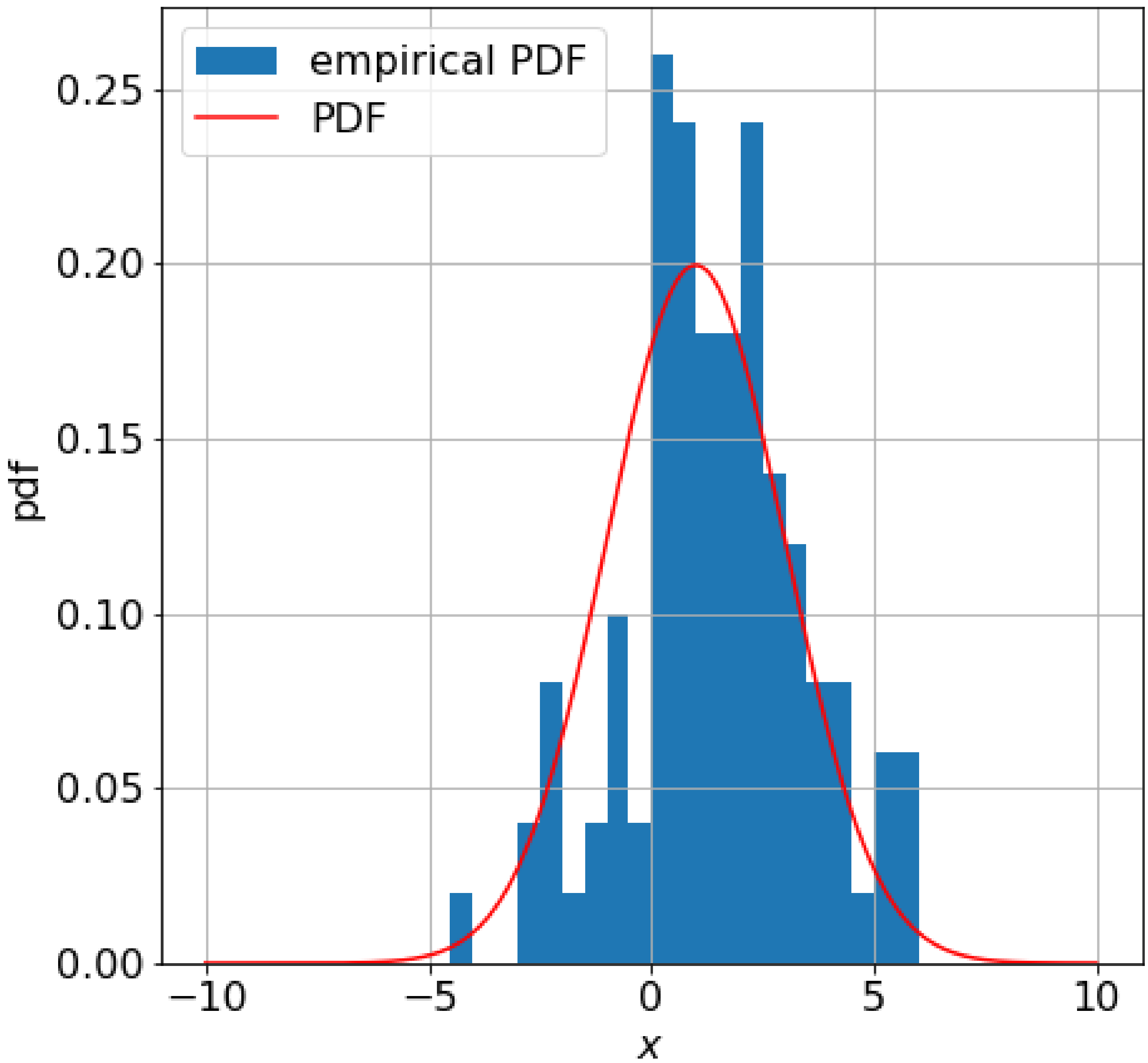
events per $x = 0.5$ bin
0
2
4
6
8
10
12
−10
−5
0
5
10
x

PDFs and Empirical PDFs

- If I make n measurements of a random variable x, the results can be sorted into a histogram to determine the experimental distribution.
- If I divide the number of events in each bin by

  (total number of events)x(bin width)

  the result is an empirical PDF that I can compare to a (model) PDF.


empirical PDF
PDF
0.25
0.20
0.15
0.10
0.05
0.00
pdf
−10
−5
0
5
10
x

PDFs and Empirical PDFs

- If I make n measurements of a random variable x, the results can be sorted into a histogram to determine the experimental distribution.
- If I divide the number of events in each bin by

  (total number of events)x(bin width)

  the result is an empirical PDF that I can compare to a (model) PDF.
- The physics lies in the descriptive statistics (mean, variance, skewness, kurtosis…) of the PDF, which we must try to infer.

Binomial Distribution

Binomial Distribution

- The probability P_B of observing k successes in N trials, where the probability of success per trial is p, is given by:

$$P_B(p, N, k) = \frac{N!}{k!(N-k)!} p^k (1-p)^{N-k}$$

- The average number of successes:

$$\overline{k} = Np$$

- The standard deviation of the number of successes σ_k is given by:

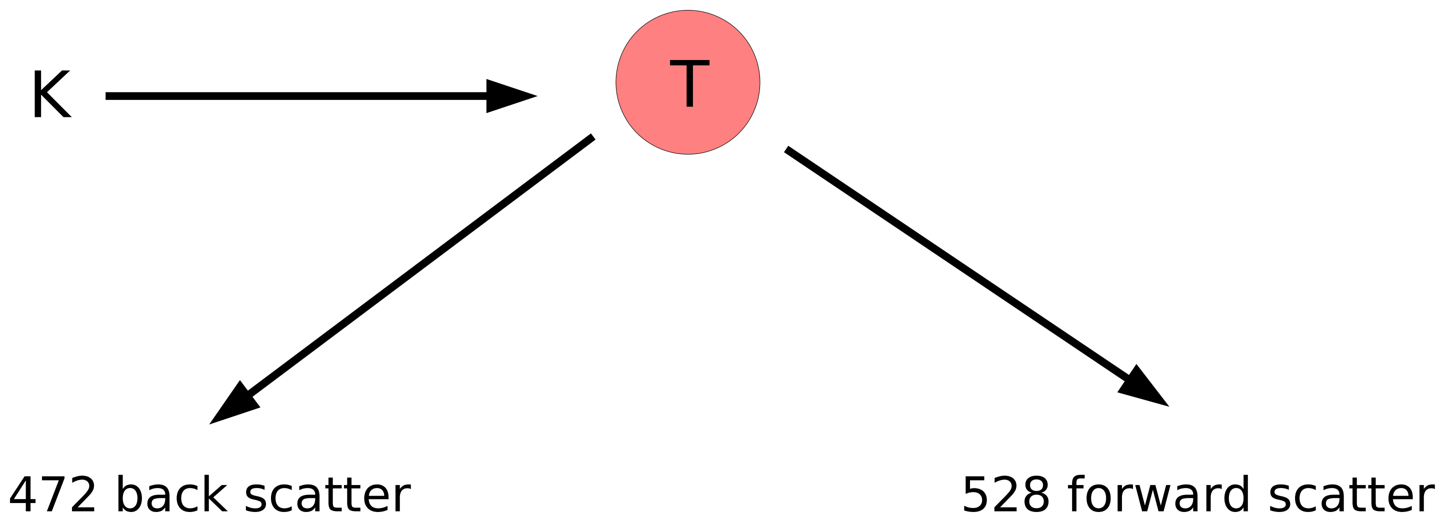
$$\sigma_k = \sqrt{Np(1-p)}$$

Example: If I toss a coin 3 times, what is the probability of obtaining 2 heads?

Example: A hospital admits four patients suffering from a disease for which the mortality rate is 80%. Find the probabilities that (a) none of the patients survives (b) exactly one survives (c) two or more survive.

Example: In a scattering experiment, I count 1,000 forward- and backward scattering events. A model predicts 50% forward and 50% backward.

What I observe:



What uncertainty should I quote? Is this consistent with the model?

Poisson and Gaussian Distributions

Binomial
Distribution

$$
\begin{aligned}
p &\longrightarrow 0 \\
N &\longrightarrow \infty \\
\text{S.T.} \quad \overline{\nu} = \mu &= \text{finite, small}
\end{aligned}
$$

Poisson
Distribution

Poisson Distribution: The probability P_P of observing n in a Poisson-distributed data set with mean μ is given by:

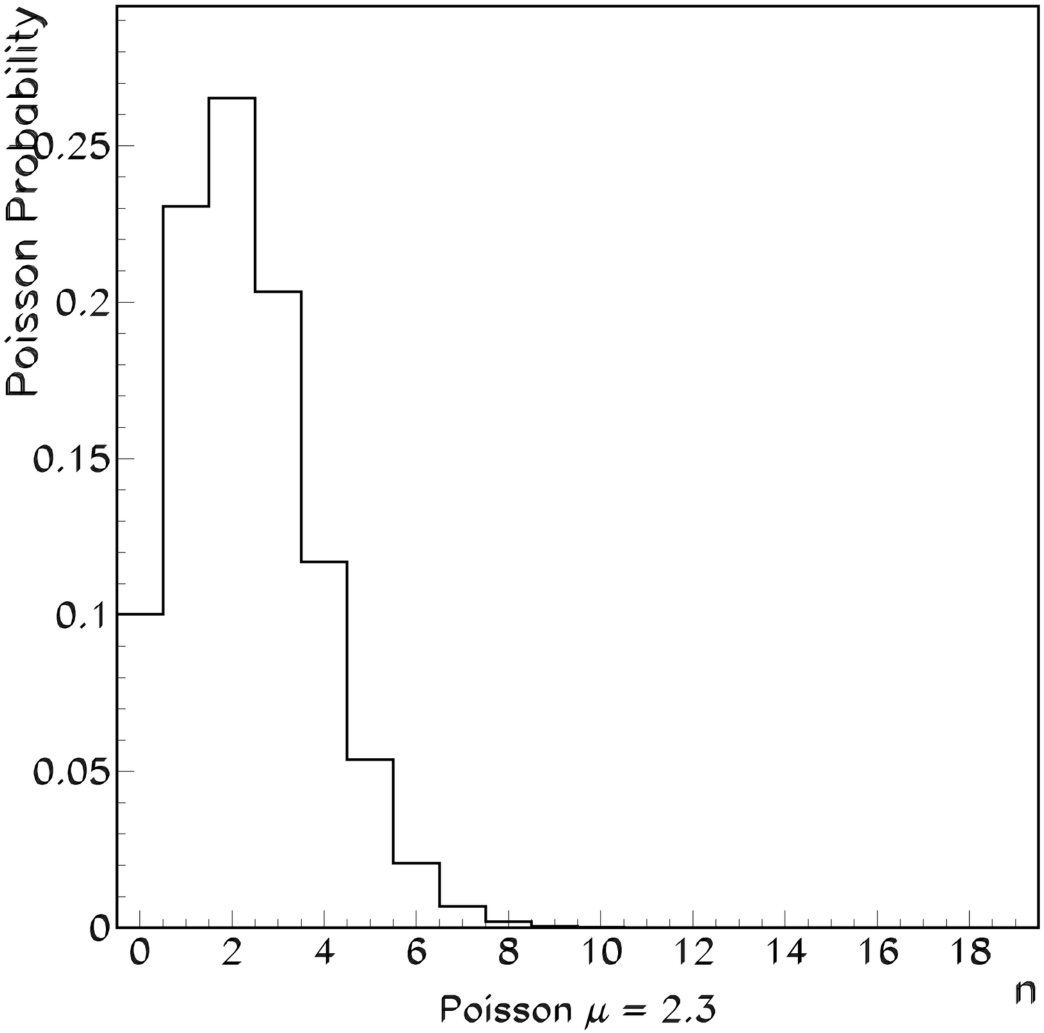
$$P_P(n;\mu) = \frac{\mu^n}{n!}e^{-\mu}$$

The standard deviation of the distribution is given by

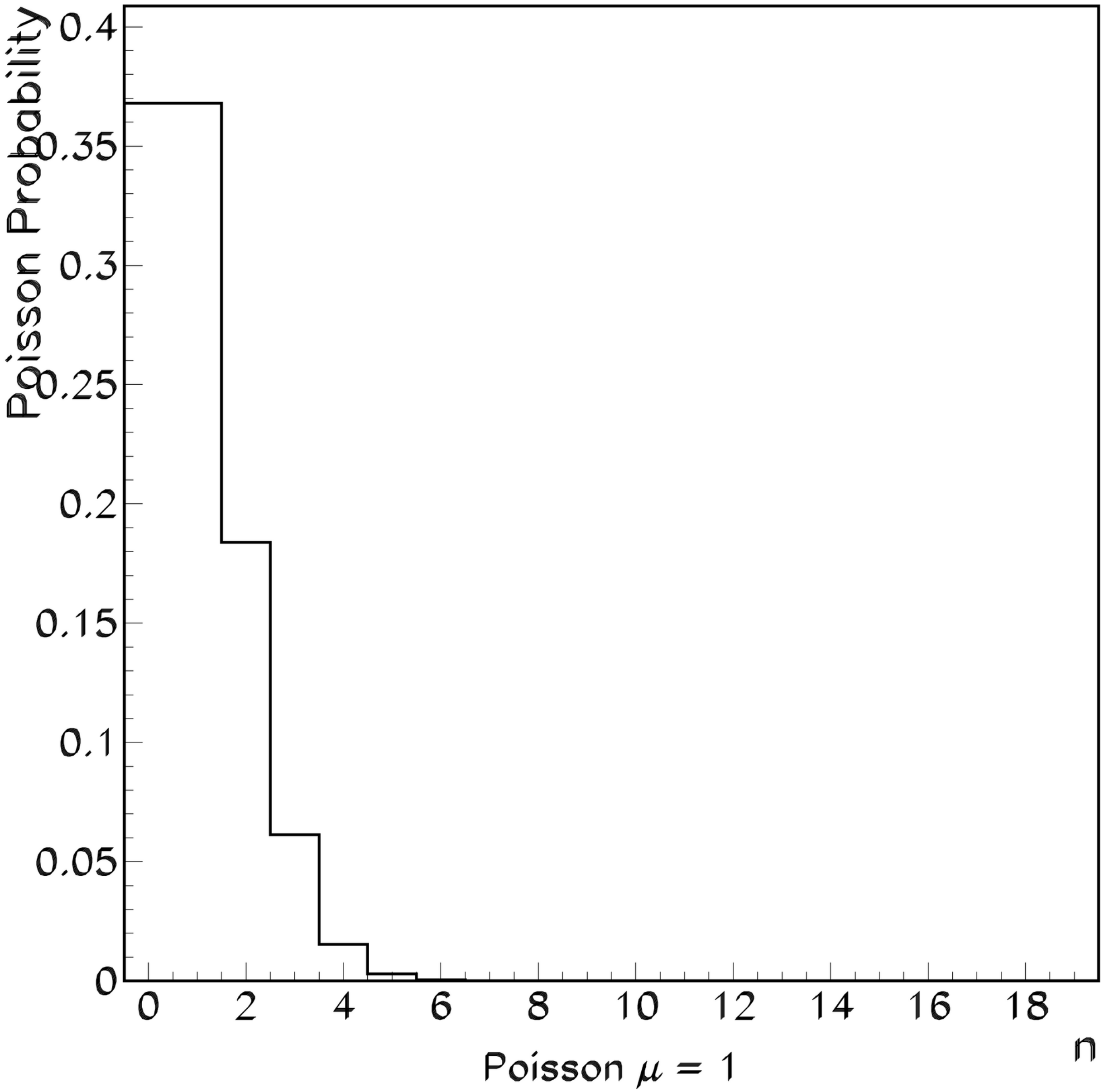
$$\sigma_n = \sqrt{\mu}$$

Example of Poisson Distribution

- Poisson distributed data can take on discrete integer values.
- n must be an integer
- μ need not be!



What Happens as μ Becomes Large?



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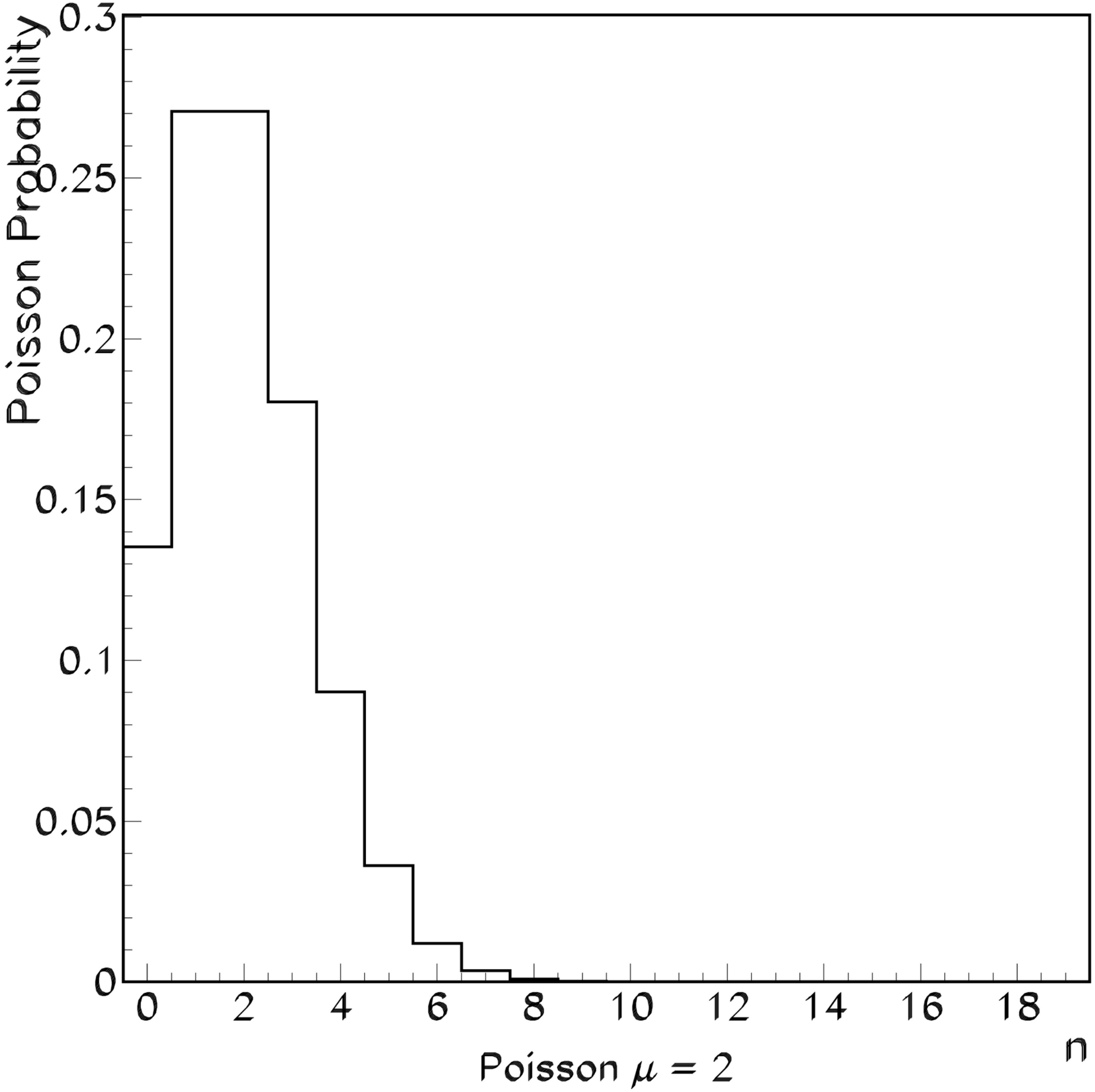

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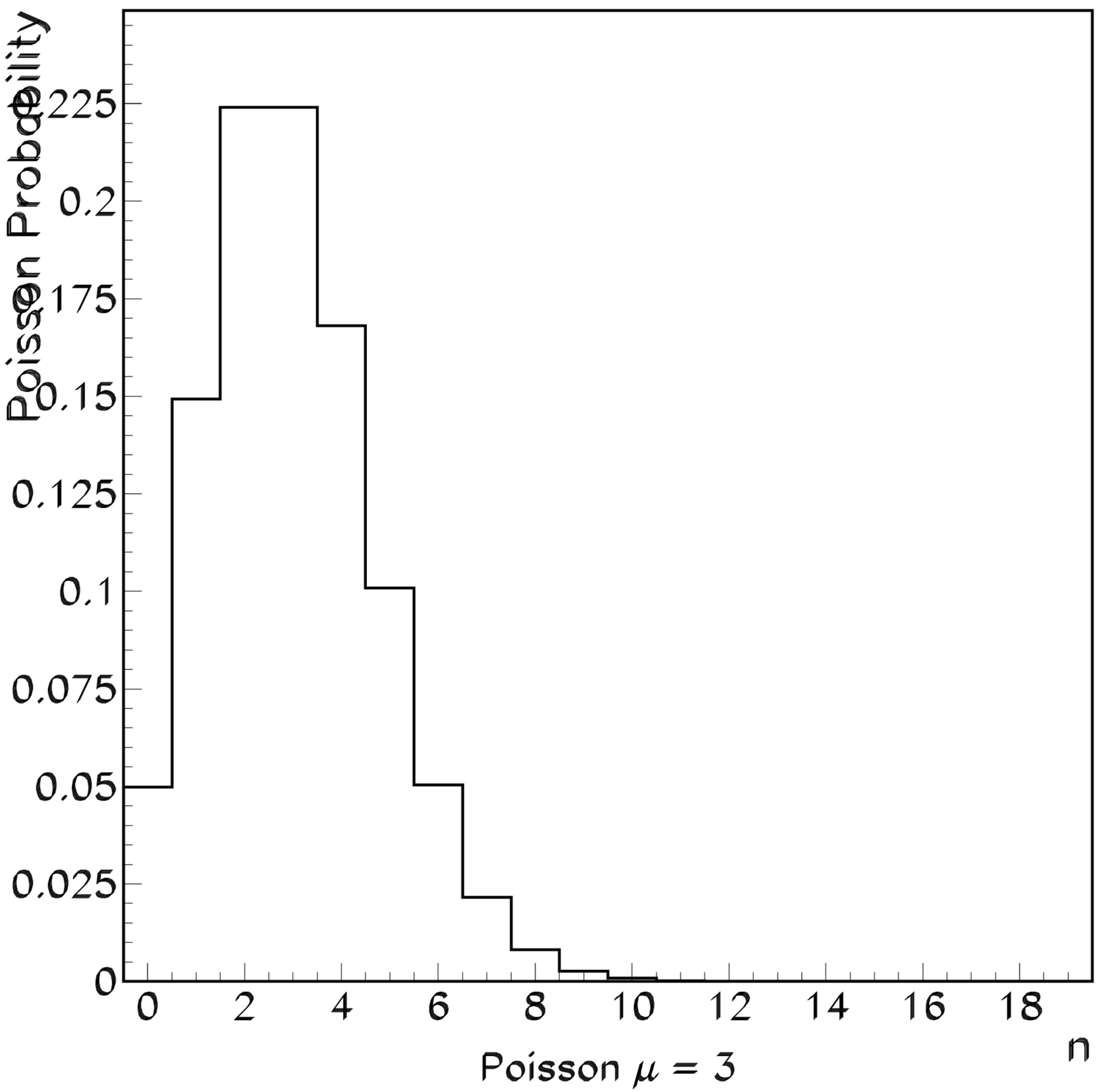

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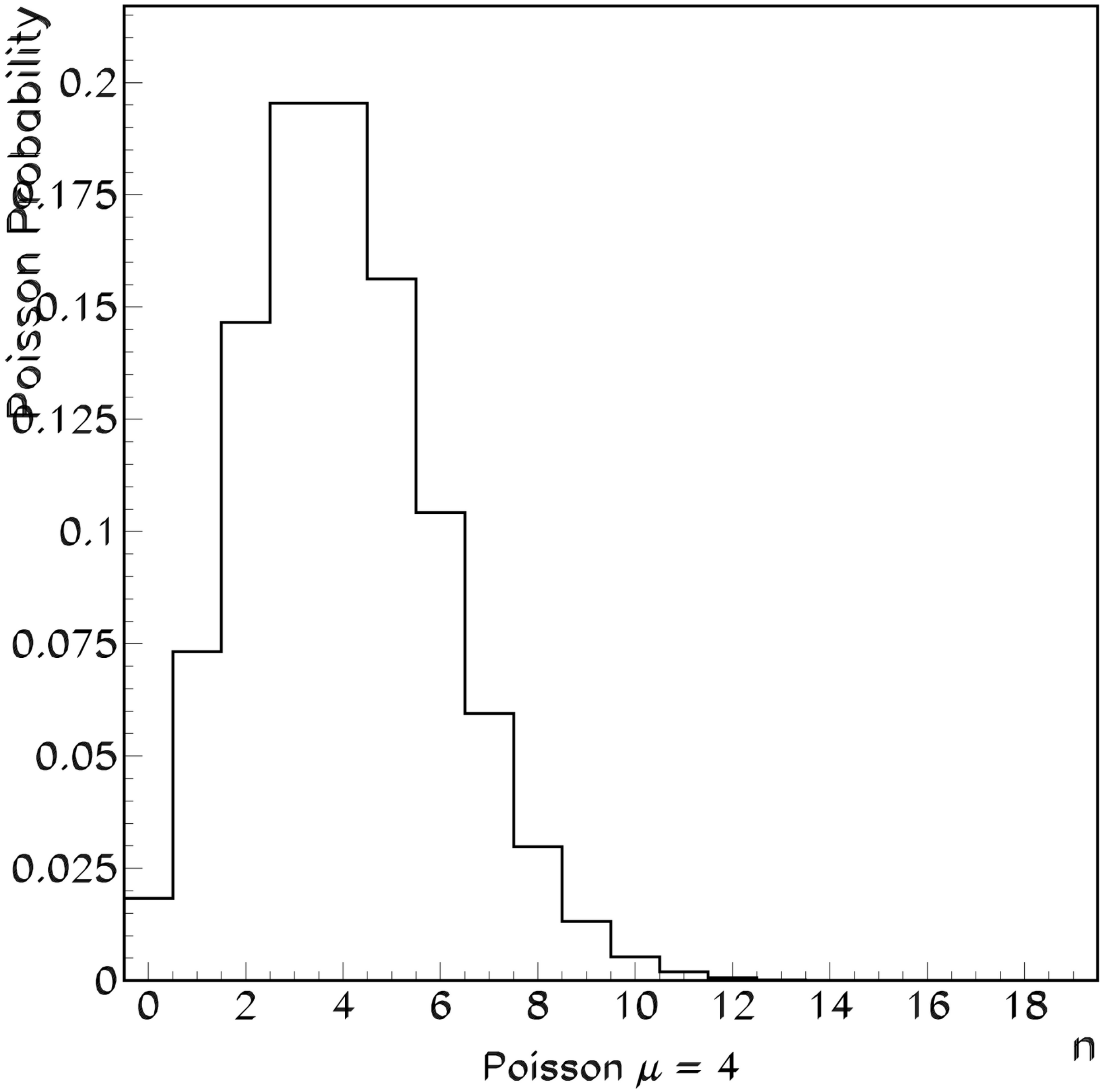

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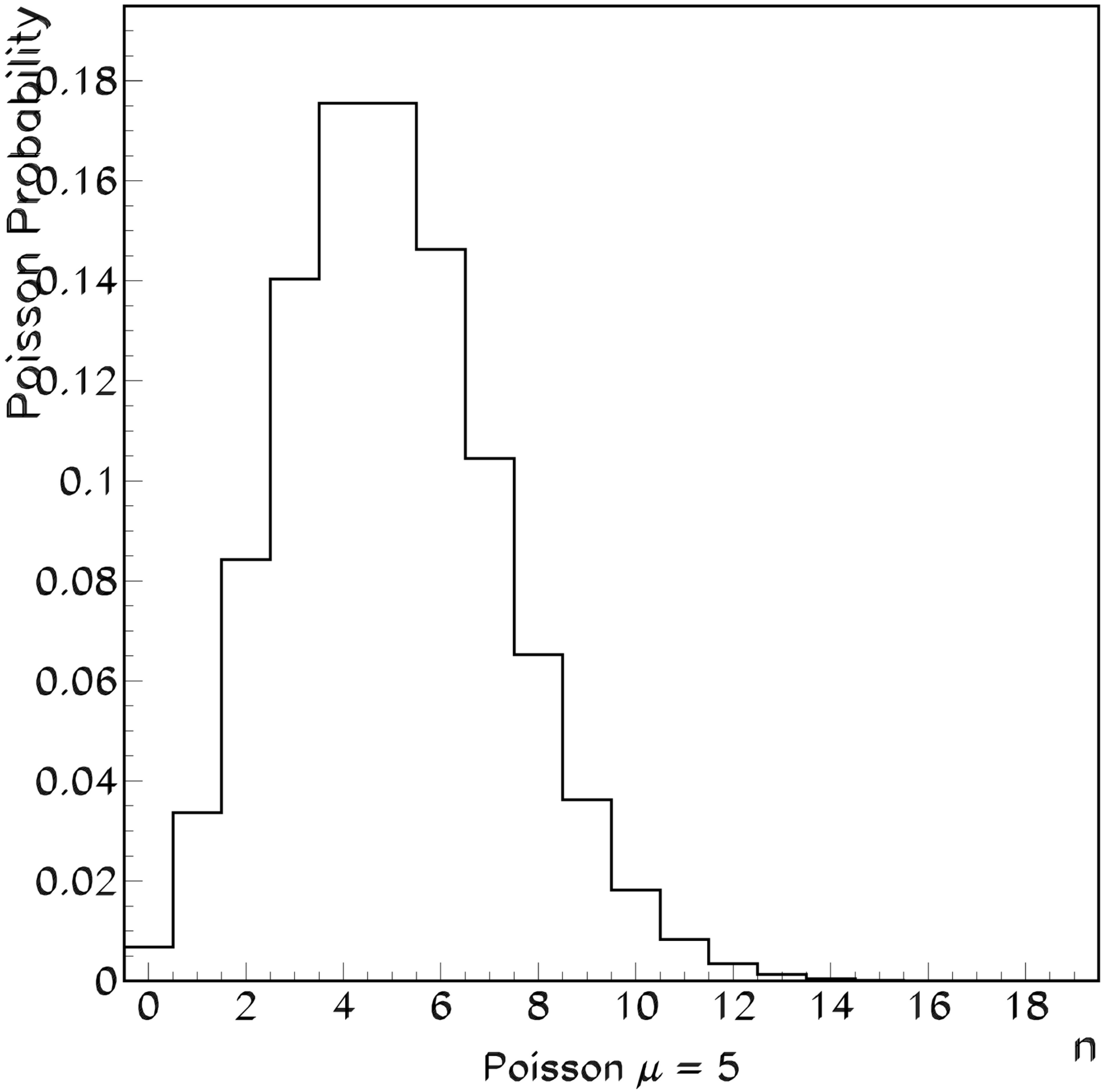

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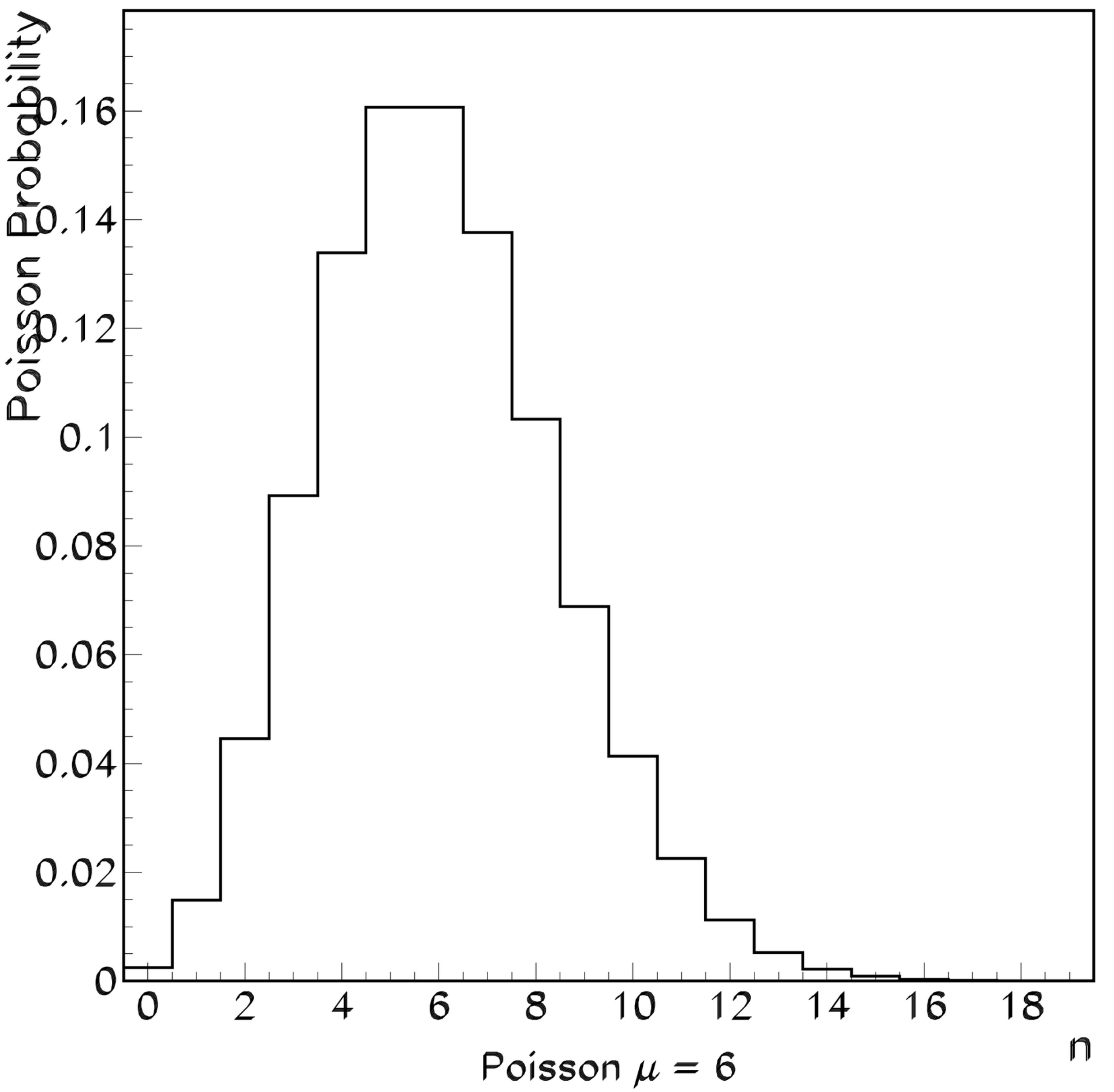

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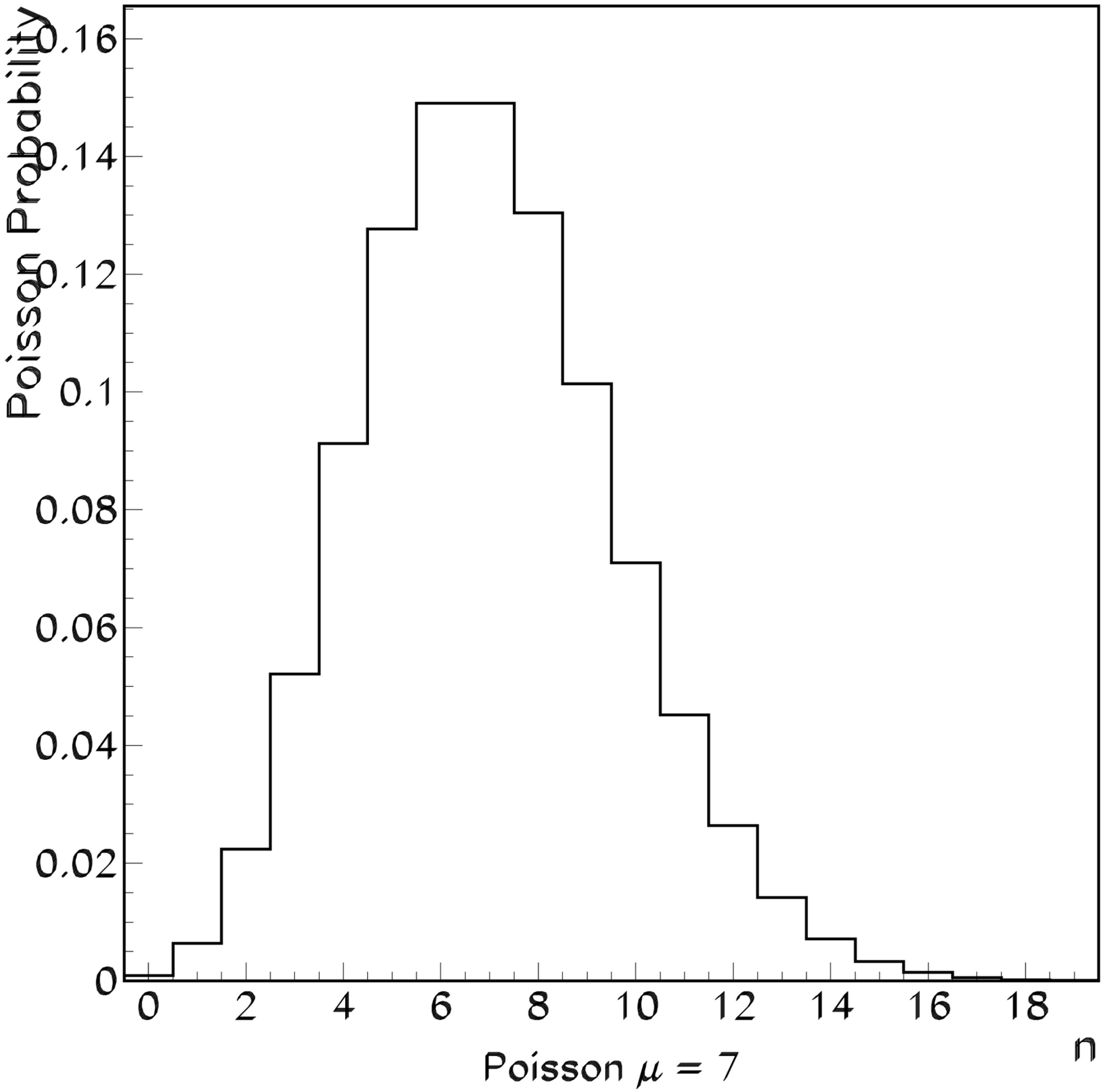

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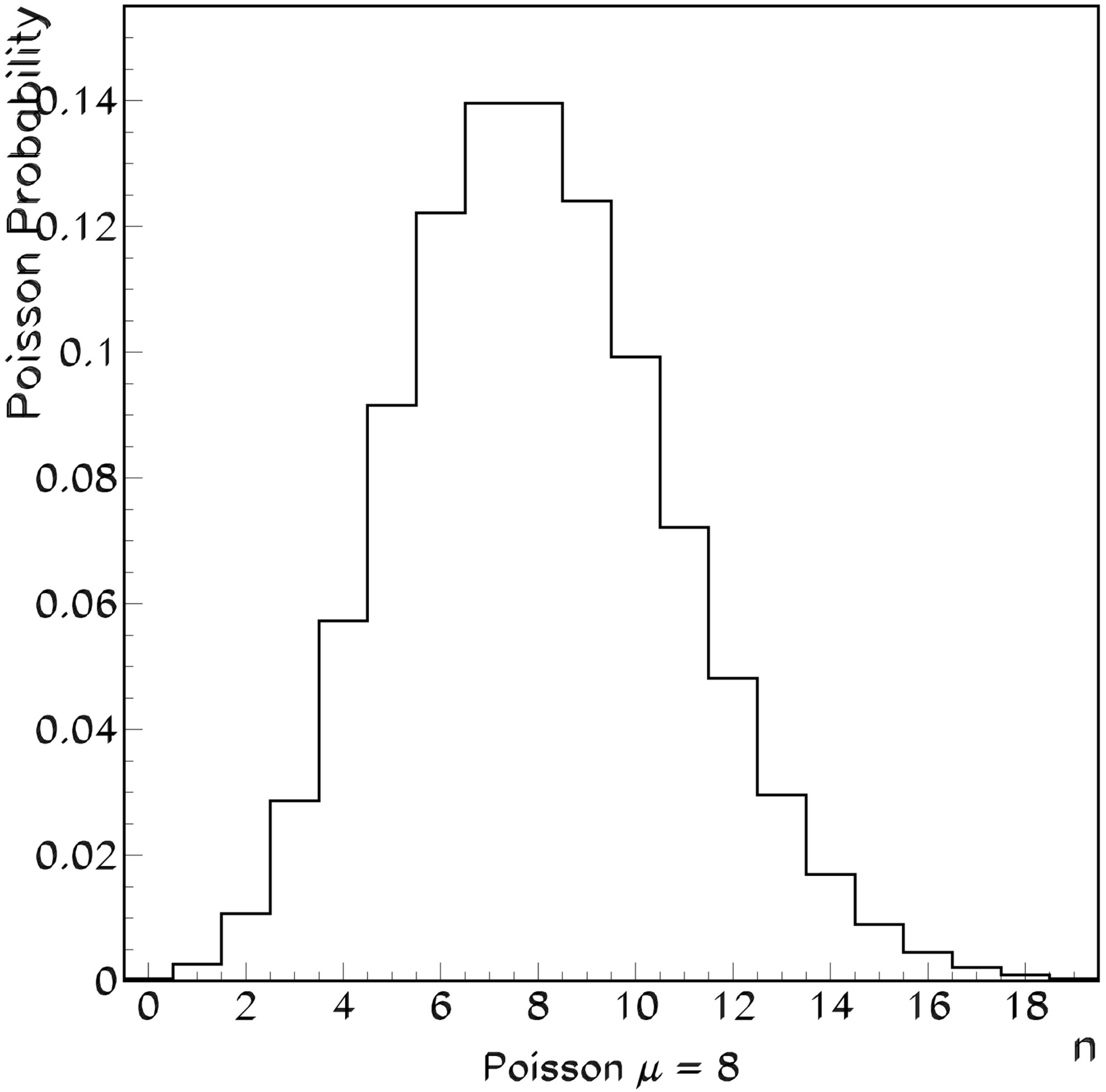

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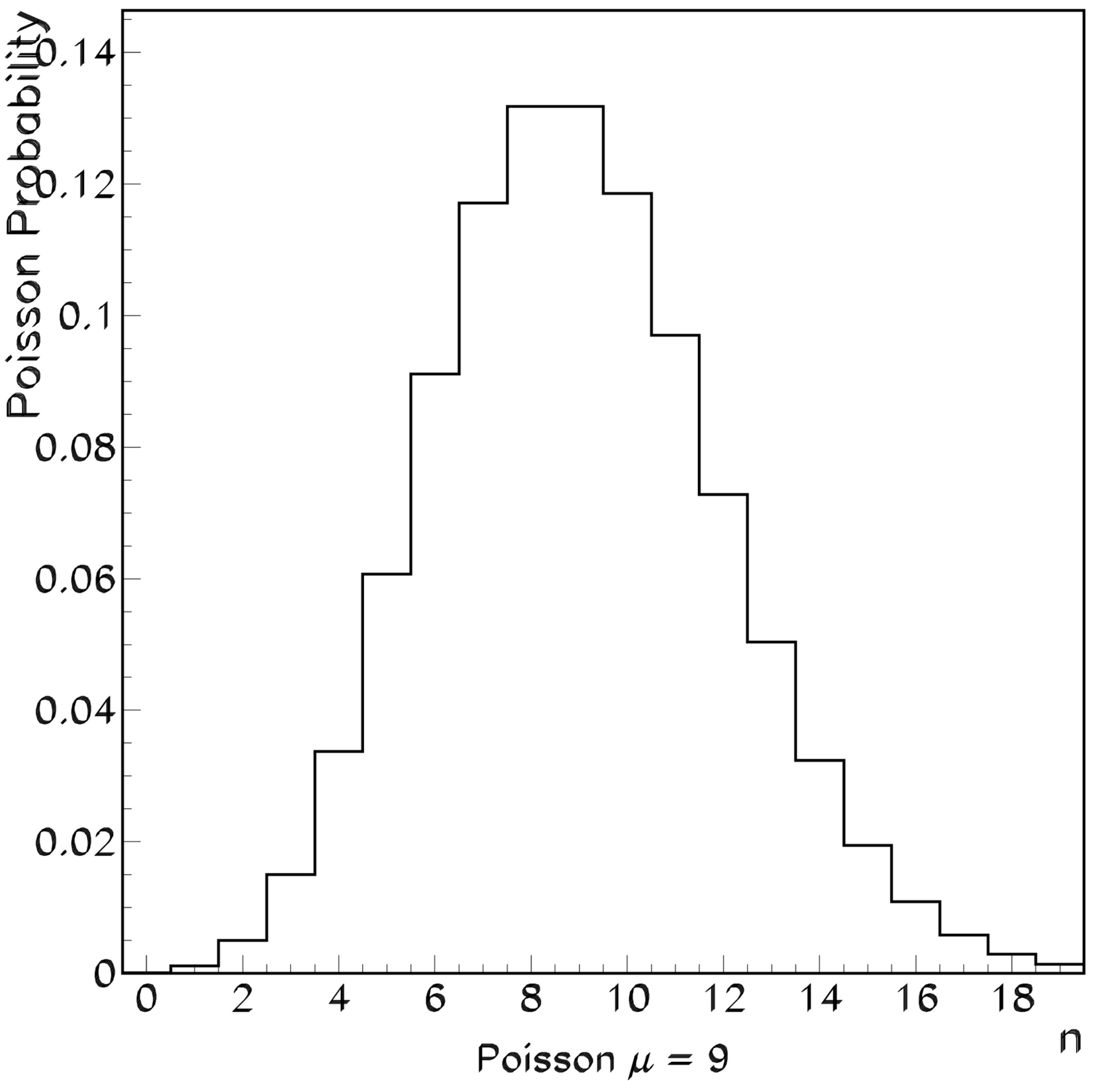

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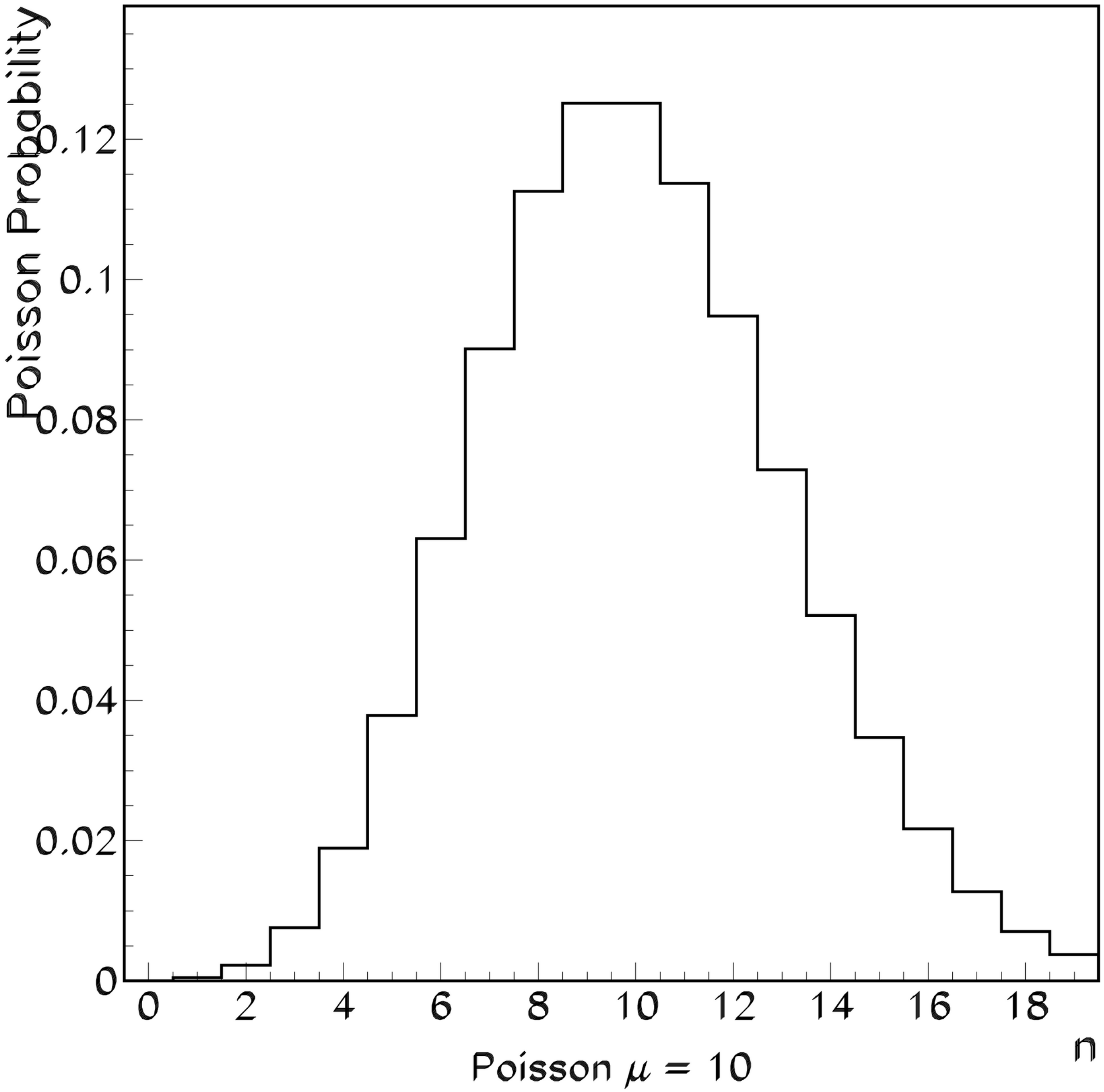

Binomial Distribution

$$p \longrightarrow 0$$
$$N \longrightarrow \infty$$
$$\text{S.T.} \quad \overline{\nu} = \mu \; = \text{finite, small}$$

Poisson Distribution

$$p \longrightarrow 0$$
$$N \longrightarrow \infty$$
$$\text{S.T.} \quad \overline{\nu} = \mu \; = \text{finite, large}$$

Gaussian (Normal) Distribution (Still Discrete!)

Gaussian (or Normal) Distribution: The probability P_G of observing n in a normally-distributed data set with mean μ is given by:

$$P_G(n;\mu) = \frac{1}{\sqrt{2\pi}\sigma_n} e^{-(n-\mu)^2/2\sigma_n^2}$$

The standard deviation of the distribution is given by

$$\sigma_n = \sqrt{\mu}$$

Explain: Using a Geiger counter, I measure the activity of a weakly radioactive rock. I record a small number (<5) counts in a ten second interval.

Why do I expect the number of counts I'd measure in repeated trials to be Poisson Distributed?

Discussion

- Can a Geiger-detector counting experiment be treated as a *binomial distribution* problem?
- What are some practical difficulties one might encounter in doing so?
- Would an interpretation via the *Poisson or Gaussian distributions* work?

Additional Reading:

Taylor: *An Introduction to Error Analysis*

Bevington: *Data Reduction and Error Analysis for the Physical Sciences*